Android Advanced Mastery

Biometrics · Google Maps · App Bundles · In-App Billing

Biometric Authentication

Secure, modern identity management

CORE COMPONENTS

Biometric Manager

Verified hardware capabilities and user settings. It ensures the device supports fingerprints or face recognition before attempting authentication.

`canAuthenticate()`

Biometric Prompt

A standard system dialog that manages the UI flow, handles failed attempts, and enforces security timeouts automatically.

`authenticate(promptInfo)`

IMPLEMENTATION WORKFLOW

Permissions

Add USE_BIOMETRIC and CAMERA to the Manifest to allow sensor access.

Availability

Check security status using BiometricManager to disable features if biometrics are missing.

Callbacks

Register success, error, and failure listeners to respond to user interactions.

Google Maps Integration

Bringing the world to your interface

SETUP REQUIREMENTS

Cloud Console

Create a project and enable the **Maps SDK for Android** and **Geocoding API**.

API Keys

Generate unique keys and secure them via package restrictions in the Play Console.

Billing

A Google Cloud billing account is required, though most individual testing falls under free tiers.

DYNAMIC MAP TYPES

Mode	Description	Use Case
NORMAL	Standard view with roads and labels.	City navigation & routing.
SATELLITE	Real-world imagery from space.	Nature exploration.
HYBRID	Satellite imagery with roads superimposed.	Context-aware navigation.
TERRAIN	Topographical data and contour lines.	Hiking and elevation tracking.

GEOCODING & MARKERS

Location Translation

Forward Geocoding: Converts text addresses into Latitude/Longitude coordinates.

Reverse Geocoding: Translates specific coordinates into human-readable street addresses.

Custom Markers

Add Marker composables to indicate specific POIs, complete with titles and snippet text for user info windows.

Android App Bundles

Optimized delivery for global users

THE BUNDLE REVOLUTION

Size Efficiency

App Bundles (.aab) reduce download sizes by including only the code and resources required for a user's specific device configuration.

Dynamic Delivery

Modules can be downloaded on-demand. Only the base code is installed initially, with extra features added as the user needs them.

RELEASE PROCESS

Keystores

Generate private keys to sign your app, uniquely identifying you as the developer.

App Signing

Opt into Google Play App Signing for secure storage of your final distribution keys.

Internal Tracks

Roll out to authorized testers before full production to catch configuration errors.

In-App Billing

Monetization strategies and implementation

PRODUCT MODELS

Managed Products

Consumable: Purchased multiple times (e.g., game currency).

Non-Consumable: One-time unlock (e.g., Ad-free mode).

Subscriptions

Recurring revenue based on renewal cycles (monthly/annual). Includes base plans, offers, and grace periods.

PURCHASE LIFECYCLE

Billing Client

Initialize the client and establish a connection to the Billing Library service.

Launch Flow

Invoke launchBillingFlow to trigger the standard Google Play checkout UI.

Acknowledgment

All purchases must be acknowledged or consumed within 3 days or they are automatically refunded.

Questions?

Advanced Android Features Summary

Biometrics Maps Bundles Billing